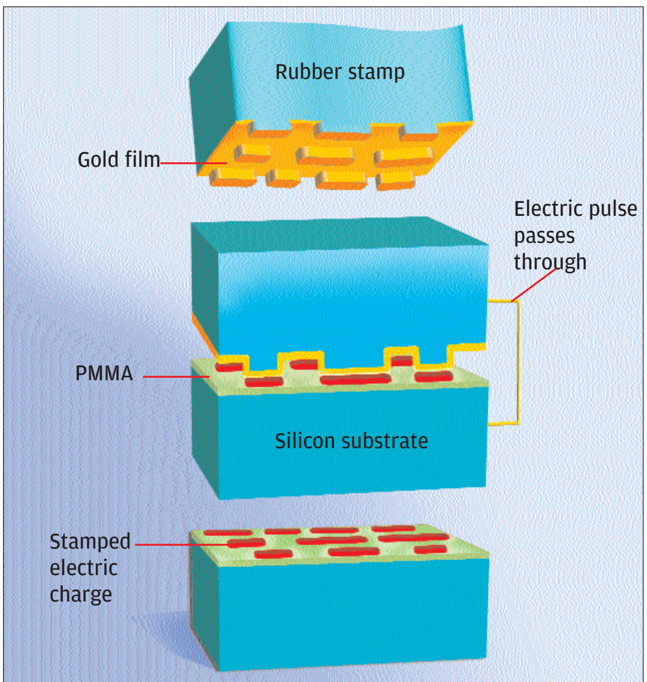
Rubber Stamps

In 2001, two Harvard University researchers designed a high-capacity data storage device based on the concept of a rubber stamp transferring patterns—1s and 0s, of course—onto a plastic film. The system could hold about 650 MB (one CD) of data on a square centimetre.

How Does It Work?

The method works on the idea of using electric charge—rather than magnetic orientations—to store bits. First, a pattern of 1s and 0s were etched on a piece of semiconductor. This piece was a mould for a liquid polymer, and solidifying the polymer resulted in the rubber stamp (see figure alongside). The 5-mm-thick stamp was then coated with an 80-nm gold film.

To write the pattern, the researchers placed the stamp on a silicon wafer covered with an 80-nm film of a substance called polymethyl methacrylate (PMMA), a polymer that can hold electric charge. They then applied an electric pulse to the gold plate. The current passed right through the PMMA film to the silicon wafer, and what resulted was an electric charge in the areas of the PMMA film that were in contact with the stamp. Where there's an electric charge, it's a 1; where it's not, it's a 0.



1. The rubber stamp with the gold film 2. The stamp in close contact with a PMMA-coated surface 3. The PMMA holds the stamped electric charge

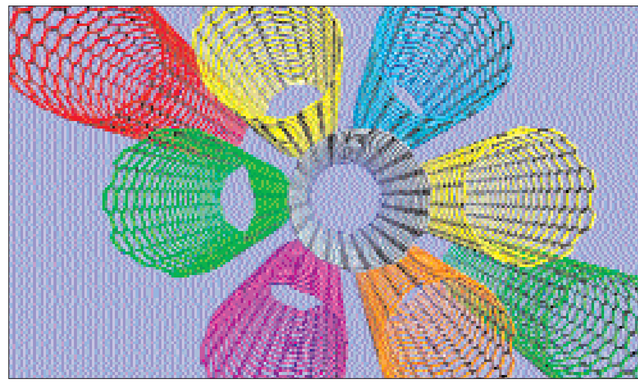
Infographics Shrikrishna Patkar

What Promise Does It Hold?

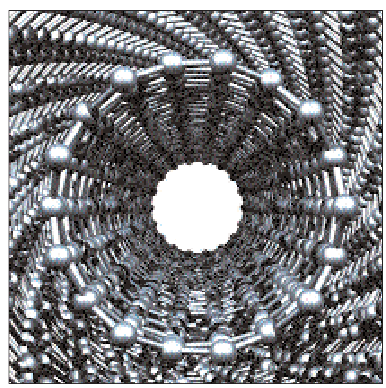
The system is low-cost, which is its main advantage. But this seems to us an exotic piece of research, and its immediate or long-term value is not clear. What is also not clear is how data would be transferred between two systems: the rubber stamp was created using photolithography and electron beam lithography, so how would data patterns be created on an everyday basis? Still, the research shows that it is possible to use electric charge storage as an alternative to magnetic storage; and that it can lead to high data densities. So OK, here's yet another nanotech hard disk alternative!

Multiwalled Carbon Nanotubes

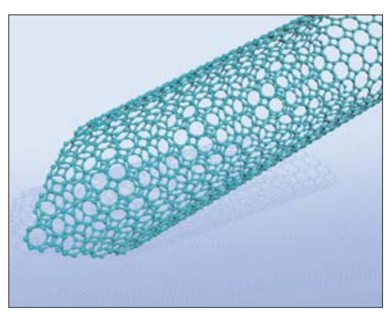
Carbon nanotubes have been much researched. They are folded sheets of carbon atoms, as in the picture alongside, where several nanotubes are shown in different colours. Multiwalled nanotubes are concentric tubes that hold together as a structure, as in the picture shown below. They're "nanotubes," so, of course, they are measured on the nanoscale—a nanotube can be smaller than a nanometre in diameter. It turns out that nanotubes can be used to write data: the tube is treated like a needle to make fine changes in a medium.



A bunch of single-walled carbon nanotubes. This isn't a photograph, but represents nanotube structure



A double-walled carbon nanotube is one tube inside another



A multi-walled nanotube can be used like a pen to write data onto a polymer

What's Being Done

Researchers from IBM Research in Zurich, Switzerland, the Japanese Nanotechnology Research Institute, and Osaka Prefecture University in Japan, have demonstrated a storage density of 250 gigabits per square inch, using the tips of multiwalled nanotubes to write bits onto a film of a certain polymer. The nanotube tip works something like a probe, pressing 1s onto the polymer surface; obviously, the absence of a 1 means a 0.

What Promise Does It Hold?

The Swiss and Japanese researchers said that using nanotube tips in practical devices could not be possible until around 2008. 250 gigabits per square inch is high, but not astronomical by any means; what *is* astronomical is the *50 million* gigabits per square inch that is being envisaged in a different scheme! That's *one crore* CDs on a square inch! Here, nanotube tips are used to place hydrogen atoms on a diamond or silicon surface.

The state of the art certainly does not match up to what is possible in theory, but you never know when a nanotech breakthrough just happens!